DOMINATED TWINS AND THE KAZHDAN–LUSZTIG CONE BEYOND HOOKS: FIRST TWINS IN S_{14}

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Corresponds to work-in-progress commit `fd667bf` (`grandpa-rick/work-in-progress`, `notes/`).
Sources: `proofs/2026-11-06-q68-kl-cone-twins.md` (PROVE 279),
`proofs/2026-11-06-q68c-s14-twin.md` (PROVE 280);
scripts `proofs/wake279/`, `proofs/prove280/`; registry `proofs/registry/seedQ2-kl-cone.json`.

Claims and registry grades (copied from `seedQ2-kl-cone.json`; nothing upgraded).	
Lemma A (dominated twin $\Rightarrow$ C_{KL} not maximal), §1	proved
Theorem B ($m(s,t) \geq 4 \Rightarrow$ twin in the cell of $w_{S\setminus\{t\}}$), §2	proved
irreducibility of those cells (B_3–B_5, H_3, F_4 census)	computed
Prop. C1 (star kill) / Cor. C2 ($\varnothing \neq L(z) \subseteq \operatorname{int} L(x)$), §3	proved
(pairing half owned by Stembridge 2008, Props. 4.4, 4.2)	
Lemma C4 (Case-B constraints), §3	proved
Theorem C3 (no twins in S_n, $n \leq 13$; computer-assisted, exact), §3	proved
($\star$) fails at S_{14}, $(4,4,3,3)$	computed
Lemma S (KL79 Thm. 4.2, $m = 3$; re-proved), §4	proved
Theorem T14 (twin in one left cell of shape $(4,4,3,3)$), §4	proved
Theorem T14′ (exactly four twin shapes in S_{14}; 14 is least), §4	proved
twin in *every* left cell of the four shapes	computed
B_3 witness; Coxeter scan; reducible non-uniqueness; GW convention check	computed
"No twins in simply-laced W"	dead-end (refuted)
Type A: twin $\Leftrightarrow$ ($\star$)-violation, §6	hunch
H_3 type 5_2 non-twin failure, §6	hunch
Q68 root (beyond type A)	in-progress

Summary. Goertzen–Williamson [GW] ask which invariant cone is the maximal one in a Specht module. They prove it is the Kazhdan–Lusztig cone C_{KL} for hooks, two-column shapes and $(n-2,2)$. They also show this fails at $(4,4,1) \vdash 9$, by an explicit non-simplicial witness. This note isolates one mechanism, a *dominated twin*, that forces C_{KL} to be non-maximal. Theorem B shows that every bond with $m \geq 4$ produces a twin, so a twin occurs in every non-simply-laced finite Coxeter group. In type A there are no twins for $n \leq 13$. The first twins are in S_{14}, in exactly four shapes. Non-maximality of C_{KL} in type A is GW's result, not ours; see §5.

1. Setup, conventions, and the twin lemma

Let W be a finite Coxeter group with simple reflections S, and let $L(w) = \{s \in S : sw < w\}$ be the left descent set. We write $\mu(x,y) = \mu(y,x)$ for the symmetrised Kazhdan–Lusztig μ-function. It is non-negative (Kazhdan–Lusztig for Weyl groups; Elias–Williamson in general). Let Γ be a left cell, and let M_Γ be the cell module at $v = 1$ with basis $\{e_x : x \in \Gamma\}$. We

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use the W-graph convention of [GW] (their convention for the KL left-cell representation; our calibration against their Ex. 4.7 and A_3 is computed):

$$(1+s)e_x = \begin{cases} 0, & s \in L(x), \\ 2e_x + \sum_{y\in\Gamma,\ s\in L(y)} \mu(x,y)\, e_y, & s \notin L(x). \end{cases}$$

Put $C_{\mathrm{KL}} = \mathrm{cone}\{e_x : x \in \Gamma\}$. A cone is *maximal* if it is maximal among pointed $\{1+s\}$-invariant cones. In type A, $W = S_n$ and $w \leftrightarrow (P(w), Q(w))$ by RSK. Left cells are $\{Q = \mathrm{const}\}$, $L(w) = \mathrm{Des}P(w)$, and $M_\Gamma \cong S^\lambda$.

Convention trap. In the C' convention, $(1+s)c_w = 2c_w$ for $sw < w$, and maximality there is equivalent to minimality for Γw_0. Maximality in that convention already fails in A_3 (computed). Do not compare C' numerics against [GW].

The following standard facts are used. (F1) [KL79, (2.3.e)]: if $y < w$, $s \in L(w) \setminus L(y)$ and $\mu(y,w) \neq 0$, then $y = sw$. (F2): if $\mu(y,w) \neq 0$ and $s \in L(y) \setminus L(w)$, then $y \leq_L w$. Consequently, if each of $L(y)$, $L(w)$ fails to contain the other and $\mu \neq 0$, then $y \sim_L w$. (F3) Subword property: if $u \leq w \in W_J$, then $L(u) \subseteq J$.

Definition 1.1. A *dominated twin* in Γ is an ordered pair $x \neq z$ in Γ such that

$$L(z) \subseteq L(x) \quad \text{and} \quad \mu(y,z) \geq \mu(y,x) \ \text{ for every } y \in \Gamma \text{ with } L(y) \not\subseteq L(x).$$

Lemma A. *If (x,z) is a dominated twin in Γ, put $v = e_z - e_x$. Then $K = \mathrm{cone}(\{e_w : w \neq z\} \cup \{v\})$ is simplicial, $\{1+s\}$-invariant, and strictly contains C_{KL}. Hence C_{KL} is not maximal.*

Proof. $\{e_w\ (w \neq z), v\}$ is a basis, so K is simplicial and hence pointed. Since $e_z = v + e_x$, we get $C_{\mathrm{KL}} \subseteq K$. Also $v \notin C_{\mathrm{KL}}$, because its e_x-coefficient is -1. For $w \neq z$ we have $(1+s)e_w \in C_{\mathrm{KL}} \subseteq K$. Now consider $(1+s)v$. (1) If $s \in L(z)$, then $s \in L(x)$, and $(1+s)v = 0$. (2) If $s \in L(x) \setminus L(z)$, then $(1+s)v = (1+s)e_z \in C_{\mathrm{KL}}$. (3) If $s \notin L(x)$, then $s \notin L(z)$, and

$$(1+s)v = 2v + \textstyle\sum_{y\in\Gamma,\, s\in L(y)} (\mu(z,y) - \mu(x,y))\, e_y.$$

Every y in the sum has $s \in L(y) \setminus L(x)$. So $L(y) \not\subseteq L(x)$, and the coefficient is ≥ 0. Moreover $y \notin \{x,z\}$, since s lies in neither $L(x)$ nor $L(z)$. Hence $(1+s)v \in K$. □

Only $\mu \geq 0$ and the displayed action are used. No Hecke algebra computation enters beyond knowing μ.

2. Every $m \geq 4$ bond produces a twin

Theorem B. *Let W be a finite Coxeter group, and let $s,t \in S$ with $m(s,t) \geq 4$. Put $J = S \setminus \{t\}$, $x = w_J$ and $z = stw_J$. Then x and z lie in one left cell Γ, and (x,z) is a dominated twin. Hence $C_{\mathrm{KL}}(\Gamma)$ is not maximal.*

Proof. Shape of x. Write $x = au$ with $a \in \langle s,t\rangle$, u minimal in its coset $\langle s,t\rangle u$, and lengths adding. Reduced words of x use only letters of $J \not\ni t$, so $a \in \{e, s\}$. Since $s \in J = L(x)$, we get $x = su$.

z is reduced and $L(z) \subseteq J$. The elements $tx = tsu$ and $z = stsu$ have lengths $\ell(u)+2$ and $\ell(u)+3$, because sts is reduced in the dihedral group of order $2m \geq 8$. In $I_2(m)$ with $m \geq 4$, we have $L(sts) = \{s\}$, so $t \notin L(z)$. Hence $L(z) \subseteq J = L(x)$, and $z \neq x$.

Same cell. $\mu(x,tx) = \mu(tx,z) = 1$, since each pair differs by one left multiplication. Now $t \in L(tx) \setminus L(x)$ and $s \in L(x) \setminus L(tx)$, so (F2) gives $x \sim_L tx$. Likewise $s \in L(z) \setminus L(tx)$ and $t \in L(tx) \setminus L(z)$ give $tx \sim_L z$.

Twin condition. Let $y \in \Gamma$ with $L(y) \not\subseteq J$ and $\mu(y,x) \neq 0$. Then $t \in L(y)$. If $y < x$, then (F3) gives $L(y) \subseteq J$, which is impossible. So $y > x$ and $t \in L(y) \setminus L(x)$, and (F1) forces $x = ty$,

that is, $y = tx$. Thus the only y to check is tx, and there $\mu(tx, z) = 1 \geq 1 = \mu(tx, x)$. For every other y, $\mu(y, x) = 0 \leq \mu(y, z)$. Lemma A applies. □

Remark 2.1. (i) The case (B_3, $\{s_1, s_2\}$ the 4-bond) is the irreducible 3-dimensional cell $\{s_3s_1,\ s_2s_3s_1,\ s_1s_2s_3s_1\}$, with descent sets $\{1,3\}, \{2\}, \{1\}$ (computed, exact). Stembridge [St, p. 365] displays the same cell $\{1\}$—$\{2\}$—$\{1,3\}$, noting $\{1\} \subset \{1,3\}$, but uses it only for a Polygon Rule argument. He has no cone statement. (ii) The dihedral middle cells are the case $J = \{s\}$. (iii) Theorem B gives non-maximality in every cell of the form $\Gamma(w_{S\setminus\{t\}})$, reducible or not. Uniqueness of $C^{\max}$ needs irreducibility. That is computed only: the B_3 cells, one of the two w_J-cells in each of B_4 and B_5, H_3 (dimension 5), and all 32 irreducible 8-dimensional cells of F_4. In reducible cells such as the $I_2(4)$ middle cell, there can be two distinct maximal pointed cones (computed), so "the" maximal cone is ill-posed there.

3. The star-operation kill and no twins for $n \leq 13$

Fix s, t with $m(s, t) = 3$, and let $D_L(s, t) = \{w : |L(w) \cap \{s, t\}| = 1\}$. Each $w \in D_L(s, t)$ has a unique *form* su, tsu, tu or stu, where u is minimal in $\langle s, t\rangle u$. The *type* of w is $L(w) \cap \{s, t\}$. The star operation $*$ is the involution $su \leftrightarrow tsu$, $tu \leftrightarrow stu$ [KL79, §4]. It swaps types, and $\mu(w, *w) = 1$.

Proposition 3.1 (C1, star kill)**.** *Let W be arbitrary, let (x, z) be a dominated twin, and let $s \in L(x)$, $t \notin L(x)$ with $m(s, t) = 3$. Then $s \notin L(z)$.*

Proof. Claim: if y has type $\{t\}$, z has type $\{s\}$ and $\mu(y, z) \neq 0$, then $z = *y$. Indeed, if $y < z$ then (F1) with $s \in L(z) \setminus L(y)$ gives $y = sz$. If $z < y$ then (F1) with $t \in L(y) \setminus L(z)$ gives $z = ty$. So $z \in \{sy, ty\}$ has type $\{s\}$, and therefore $z = *y$.

Now put $x' = *x$. By (F2), $x' \in \Gamma$. Since $t \in L(x') \setminus L(x)$, the twin condition gives $\mu(x', z) \geq \mu(x', x) = 1$. Also $t \notin L(z) \subseteq L(x)$. If $s \in L(z)$, then z has type $\{s\}$ and x' has type $\{t\}$, so the Claim gives $z = *x' = x$, a contradiction. □

Attribution. The pairing step in the Claim says that, for $m(s, t) = 3$, a type-$\{t\}$ vertex and a type-$\{s\}$ vertex joined by an edge are star partners with $\mu = 1$. This is Stembridge's: the Bonding Rule [St, Prop. 4.4, $p = 3$ clause, pp. 358–359] together with the Simplicity Rule [St, Prop. 4.2, p. 357] (incomparable τ-sets give a simple edge of weight 1); see [St, Rem. 4.3] for the simply-laced case. That exactly one of sy, ty has type $\{s\}$ is classical [KL79, §4]. Applying this to twins is ours.

Corollary 3.2 (C2)**.** *Let W be simply laced, and put* $\operatorname{int} L(x) = \{r \in L(x) :$ *every neighbour of r lies in $L(x)$}. If (x, z) is a dominated twin, then $\varnothing \neq L(z) \subseteq \operatorname{int} L(x)$. Hence twins cannot occur in any cell satisfying*

$$(\star) \qquad \text{there are no } x \neq z \in \Gamma \text{ with } \varnothing \neq L(z) \subseteq \operatorname{int} L(x).$$

Here $L(z) \neq \varnothing$ because $|\Gamma| \geq 2$ excludes $z = e$.

Proposition 3.3 (C4, Case-B constraints)**.** *Let W be simply laced, let (x, z) be a twin, and let (s, t) be a boundary pair of x, meaning $s \in L(x)$, $t \notin L(x)$ and $m = 3$. Put $x' = *x$. Then either $z = tx'$, which forces $x = stu$, $x' = tu$, $z = u$; or $x' < z$ and every $r \in L(z) \setminus L(x')$ satisfies $x' = rz$. If x has at least two boundary pairs, only the second alternative occurs, and $\ell(z) \geq \ell(x)$.*

Proof sketch. $\mu(x', z) \geq 1$ and $s, t \notin L(z)$ by C2. If $z < x'$, then (F1) with $t \in L(x') \setminus L(z)$ gives $z = tx'$; since $z \neq x$ this forces $x' = sx$, $x = stu$. If $x' < z$, then (F1) applies for each $r \in L(z) \setminus L(x')$. With two pairs, the first alternative fixes $\ell(z) = \ell(x) - 2$, which contradicts $\ell(z) > \ell(x') \geq \ell(x) - 1$ for the other pair. □

Theorem 3.4 (C3)**.** *For $n \le 13$, $(\star)$ holds in every left cell of S_n. Hence S_n has no dominated twin for $n \le 13$.*

Proof. In a cell of shape λ, the descent sets that occur are $\{\mathrm{Des}(T) : T \in SYT(\lambda)\}$. By standardisation, $\#\{T : \mathrm{Des}T \subseteq B\} = K_{\lambda,\mathrm{comp}(B)}$, and by transposition, $\#\{T : \mathrm{Des}T \supseteq A\} = K_{\lambda',\mathrm{comp}(A^c)}$. Since int is monotone, $(\star)$ fails for some cell iff there exist $A \subseteq [n-1]$ and λ with $K_{\lambda',\mathrm{comp}(A^c)} > 0$ and $K_{\lambda,\mathrm{comp}(\mathrm{int}\,A)} > 0$. The cases $A = \varnothing$ (then $L(z) = \varnothing$) and $A = [n-1]$ (only $\lambda = (1^n)$, which has one tableau) are excluded. For other A, $\mathrm{int}\,A \subsetneq A$, so $z \neq x$ automatically. Since $K_{\mu,\nu} > 0 \iff \mu \trianglerighteq \mathrm{sort}(\nu)$ and conjugation reverses dominance, such a λ exists iff $\beta \trianglelefteq \alpha'$, where $\alpha = \mathrm{sort}\,\mathrm{comp}(\mathrm{int}\,A)$ and $\beta = \mathrm{sort}\,\mathrm{comp}(A^c)$. An exhaustive exact-integer check over all $A \subseteq [n-1]$ (`dominance_star.py`) finds no violation for $n \le 13$. Direct SYT enumeration agrees for $n \le 11$ (`syt_interior.py`). □

Theorem C3 rests on the computer check plus textbook RSK and Kostka facts. In QSym language, $(\star)$ fails for λ iff $s_\lambda = \sum_T F_{\mathrm{Des}T}$ contains F_A and F_B with $\varnothing \neq B \subseteq \mathrm{int}\,A$. Also computed: $(\star)$ holds in all cells of D_4 and D_5, and twin scans give 0 twins in A_3–A_6, D_4 and D_5. Controls: B_4 has twins in 24/50 cells and F_4 in 44/72.

4. The S_{14} twins

At $n = 14$, $(\star)$ fails for $\lambda = (4,4,3,3)$:

$$P = \begin{smallmatrix} 1 & 2 & 6 & 10 \\ 3 & 7 & 11 & 14 \\ 4 & 8 & 12 & \\ 5 & 9 & 13 & \end{smallmatrix}, \qquad \mathrm{Des}P = \{2,3,4,6,7,8,10,11,12\};$$

$$P' = \begin{smallmatrix} 1 & 2 & 3 & 7 \\ 4 & 5 & 6 & 11 \\ 8 & 9 & 10 & \\ 12 & 13 & 14 & \end{smallmatrix}, \qquad \mathrm{Des}P' = \{3,7,11\} = \mathrm{int}\,\mathrm{Des}P.$$

So C1/C2 alone cannot rule out twins here, and μ inside a 12012-dimensional cell has to be decided.

Theorem 4.1 (T14)**.** *Let $\Gamma \subset S_{14}$ be the left cell with $Q = \begin{smallmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 14 \\ 8 & 9 & 10 & \\ 11 & 12 & 13 & \end{smallmatrix}$. Put $x = (P,Q) = [5,9,13,14,4,8,12,3,7,11,1,2,6,10]$, of length 55, and $z = (P',Q)$, of length 67. Then (x,z) is a dominated twin. Hence $C_{\mathrm{KL}}(S^{(4,4,3,3)})$ is not maximal, and $\mathrm{cone}(\{e_w : w \neq z\} \cup \{e_z - e_x\})$ is an explicit simplicial invariant cone strictly containing it.*

Theorem 4.2 (T14′)**.** *The $(\star)$-violating pairs $(\lambda, A = L(x))$ in S_{14} are exactly the following four. Each has a unique candidate pair (P_x, P_z), and each is a dominated twin: For $(4,4,3,3)$ and $(5,3,3,3)$, $A = \{2,3,4,6,7,8,10,11,12\}$ and $\mathrm{int}\,A = \{3,7,11\}$. For $(4,4,4,2)$ and $(4,4,4,1,1)$, $A = \{1,2,4,5,6,8,9,10,12,13\}$ and $\mathrm{int}\,A = \{1,5,9,13\}$. Rows are separated by /:*

λ	P_x	P_z
$(4,4,3,3)$	1 2 6 10/3 7 11 14/4 8 12/5 9 13	1 2 3 7/4 5 6 11/8 9 10/12 13 14
$(5,3,3,3)$	1 2 6 10 14/3 7 11/4 8 12/5 9 13	1 2 3 7 11/4 5 6/8 9 10/12 13 14
$(4,4,4,2)$	1 4 8 12/2 5 9 13/3 6 10 14/7 11	1 3 4 5/2 7 8 9/6 11 12 13/10 14
$(4,4,4,1,1)$	1 4 8 12/2 5 9 13/3 6 10 14/7/11	1 3 4 5/2 7 8 9/6 11 12 13/10/14

By C2 these are all the twins of S_{14} (by P-tableaux), and by Theorem 3.4, 14 is the least n for which S_n has a twin. The shapes form two conjugate pairs. Since L-inclusion is reversed by $w \mapsto ww_0$, the conjugate twins are new pairs, not formal images.

Lemma S. The lever is the following lemma, which is [KL79, Thm. 4.2] in the case $m = 3$. KL79 is not on disk, so we re-proved it from the recursion. Write $\tilde{\mu}(a,b)$ for μ of the comparable pair, and 0 if a, b are incomparable.

Lemma S. *Let $m(s,t) = 3$. For all $y, w \in D_L(s,t)$, $\tilde{\mu}(y,w) = \tilde{\mu}(*y, *w)$.*

Proof. If y, w lie in the same coset, check the six pairs directly. Otherwise use the KL recursion [KL79, (2.2.c)]: for $tv > v$,

$$P_{x,tv} = q^{1-c}P_{tx,v} + q^c P_{x,v} - \textstyle\sum_{z<v,\ tz<z} \mu(z,v) q^{(\ell(tv)-\ell(z))/2} P_{x,z}, \qquad c = [tx < x],$$

together with (2.3.g) ($P_{x,w} = P_{tx,w}$ for $t \in L(w)$) and (F1).

Different types. The Claim in C1 gives $\tilde{\mu}(y,w) \neq 0 \Rightarrow w = *y$, and the same holds after applying $*$.

Same form (S-i), say $\tilde{\mu}(su, su') = \tilde{\mu}(tsu, tsu')$, with $d = \ell(u') - \ell(u)$ odd. Apply the recursion with $v = su'$ and $x = tsu$, and read off the coefficient of $q^{(d-1)/2}$. The first term gives $\mu(su, su')$. The second term, via (2.3.g), contributes $\mu(stsu, su')$ when $stsu < su'$. In the sum, (F1) forces $z = stsu$ with $P_{tsu,z} = 1$ and exponent 0, which contributes $-\mu(stsu, su')$. The last two cancel.

Mixed form (S-ii, gap-changing), $\tilde{\mu}(su, stu') = \tilde{\mu}(tsu, tu')$, with $\delta = \ell(u') - \ell(u) \geq 0$ even. Apply the recursion with $v = tu'$ and $x = su$, and read off the coefficient of $q^{\delta/2}$. The first term $P_{u,tu'} = P_{tu,tu'}$ has gap δ and degree $< \delta/2$, so it contributes 0. The second term $P_{su,tu'} = P_{tsu,tu'}$ contributes $\mu(tsu, tu')$. In the sum, (F1) would force $z = tsu$, but $s \notin L(tsu)$, so the sum contributes 0. For $\delta = 0$ both sides vanish (mirror computation). For $\delta < 0$, run the $s \leftrightarrow t$ mirror. The remaining cases follow by swapping s, t or reading right to left. □

Machine control (computed): $\tilde{\mu}(y,w) = \tilde{\mu}(*y, *w)$ for all 940,800 pairs in $D_L(s_i, s_{i+1})$ in S_5 and S_6, against full KL, with 0 violations. The gap-changing S-ii step is essential. Same-form steps preserve the length gap, and every certificate below uses 5–9 mixed steps.

How the certificates work. *Which y to check.* Let $A = L(x)$. Only y with $\mu(y,x) \neq 0$ matter. If $y > x$, or if $y < x$ with $A \not\subseteq L(y)$, then (F1) gives $y = rx$ for a single r. The case $y < x$, $L(y) \supsetneq A$ is impossible in shape $(4,4,3,3)$. Indeed, $\mathrm{Des} \supseteq \{2,3,4\}$ forces $2,3,4,5$ into rows 1–4, and similarly for 6–9 and 10–13, so a further descent would need a fifth row. Enumerating all 12012 SYT agrees. So the y to check are the $y = rx \in \Gamma$ with $L(y) \not\subseteq A$. For this Q they are the four star images $x'_r = s_r x$, $r \in \{2,5,9,12\}$, each with $\mu(x'_r, x) = 1$.

Certificates. For each r, apply simultaneous star operations $*_j$ (with respect to $\{s_j, s_{j+1}\}$) to the pair (x'_r, z) along an explicit path, checking at every step that both entries lie in $D_L(s_j, s_{j+1})$. By Lemma S, $\tilde{\mu}$ is constant along the path. The path ends at a Bruhat-comparable pair with length gap 1, where $\tilde{\mu} = 1$:

r	path $j_1, j_2, \ldots$	end lengths
2	$2,4,2,6,8,6,4,7,6,10,8$	$59, 60$
5	$3,6,8,10,12,10,8,7,5$	$59, 60$
9	$7,5,3,1,3,4,6,10,8$	$59, 60$
12	$11,9,7,5,3,5,6,8,6,5,11$	$61, 62$

The initial gaps $\ell(z) - \ell(x'_r)$ are 11–13. So $\mu(x'_r, z) = 1 = \mu(x'_r, x)$ for all four, and Lemma A applies. The certificates (`prove280/certs_1..4.json`) are replayed by an independent checker (`verify.py`) that uses its own RSK, its own star map, and Bruhat order via the rank-matrix criterion rather than the tableau criterion used in the search. All four shapes pass. For the conjugate pair of shapes there are three boundary y each ($r = 3, 7, 11$). The search procedure (star orbit plus parity, Bruhat and F1 kills) was validated against full KL on all 3906 within-cell pairs of S_5 and S_6. It returned both verdicts ($\mu = 1$ and $\mu = 0$), with 0 wrong and 0 undecided.

Every left cell (computed). The theorem needs only one cell. As a consistency check, the same decision was run in every left cell of each shape. In each cell, all boundary pairs gave $\mu = 1$, so the twin holds: $(4,4,3,3)$ in 12012/12012 cells, $(4,4,4,2)$ in 12012/12012, $(4,4,4,1,1)$

in 15015/15015, and $(5,3,3,3)$ in 15015/15015 (`out_allcells_*.txt`). A proof for all Q would combine right star operations with dual-equivalence connectivity of $SYT(\lambda)$. We only recall that connectivity, so it is not graded.

5. Relation to Goertzen–Williamson

What [GW] prove and state, verified against the arXiv v1 text:

- **Thm. 4.5:** "The operators $\{1+s \mid s \in S\}$ preserve a unique maximal invariant cone $C^{\max}$ and a unique minimal invariant cone $C^{\min}$ inside S^λ. Moreover, these cones satisfy the duality relation $(C^{\max})^* = C^{\min}$." Uniqueness is deduced from irreducibility via their reference [41] (Mejstrik–Protasov). Rem. 4.4 adds: "if C is the cone generated by a basis B, then C^* is the cone generated by the dual basis B^* with respect to the form."
- **Question 4.6** (open): "What are the maximal and minimal $1+s$, $s \in S$ invariant cones inside the Specht module S^λ?"
- **Thms. 4.11, 4.14, 4.17:** the KL basis spans $C^{\max}$ for hooks, two-column shapes, and $(n-2,2)$.
- **Ex. 4.21 and §1.1:** for $(4,4,1) \vdash 9$ (dimension 84), their strategy fails to verify the minimal cone property for the dual KL basis. They exhibit the invariant cone spanned by $w_1, \ldots, w_{13}, w_{15}, \ldots, w_{84}$ and $w_{14}+w_{44}$, $w_{14}+w_{84}$; in §1.1 they say the maximal cone is "no longer simplicial". The conjugate $(3,2,2,2)$ passes. They say "this is indeed the first case" and "all other tested examples in which the minimal cone property fails come from partitions, where the corresponding Young diagram contains the Young diagram of the partition $(4,4,1)$." Rem. 4.18 gives the tested range: all $\lambda \vdash n \le 10$ not containing $(4,4,1)$, plus $(3,3,3,3,3) \vdash 15$.

So **[GW] own "C_{KL} is not always maximal in type A"**, first at $n=9$. We do not claim first non-maximality at $n=14$. Our type-A claim is narrower: *there are no dominated twins for $n \le 13$, and the first ones are the four S_{14} shapes.* By Theorem C3, GW's $(4,4,1)$ failure is a non-twin mechanism. A literature grep of [GW] finds no "twin", "domin*", "quasisymmetric" or $n=14$ shapes.

Do our twins give the failure GW measure? GW's test is the minimal cone property of the dual KL basis. Every invariant pointed cone lies in a maximal one, so by the uniqueness in Thm. 4.5, $C^{\max}$ contains every invariant pointed cone. A twin gives an invariant simplicial $K \supsetneq C_{\mathrm{KL}}$, hence $C^{\max} \supsetneq C_{\mathrm{KL}}$. Dualising with Rem. 4.4 gives $C^{\min} = (C^{\max})^* \subseteq K^* \subsetneq C^*_{\mathrm{KL}}$, where C^*_{KL} is the cone of the dual KL basis. So the minimal cone property fails for the dual KL basis at all four S_{14} shapes. We think this identification is literally supported by Thm. 4.5 and Rem. 4.4, given two things: (a) our convention-C cell basis is GW's KL basis (computed calibration, not a quoted identity), and (b) Thm. 4.5 itself, whose uniqueness input we have not read first-hand. One caveat in the other direction: GW's phrase "tested examples in which the minimal cone property fails" may mean examples where *their strategy* fails to verify the property, which is weaker than exhibiting a smaller cone. We cannot tell from the text.

Observation. $(5,3,3,3)$ does not contain $(4,4,1)$, because its second row is $3<4$. The other three shapes do. So at $(5,3,3,3)$, C_{KL} is non-maximal in a shape outside GW's empirically observed $(4,4,1)$ pattern. This does not contradict [GW]: $n=14$ lies outside their tested range. Also, unlike $(4,4,1)$ versus $(3,2,2,2)$, here both members of each conjugate pair fail ($(5,3,3,3)' = (4,4,4,1,1)$ and $(4,4,3,3)' = (4,4,4,2)$).

Question 5.1. Is the twin cone of Lemma A ever equal to $C^{\max}$, or is $C^{\max}$ at the S_{14} shapes also non-simplicial, as at $(4,4,1)$? Is GW's "contains $(4,4,1)$" pattern a statement about the $(4,4,1)$ mechanism only? Our $(5,3,3,3)$ case suggests it should be read that way.

6. Open problems and asks

(1) **Hunch** (hunch): in type A, a twin exists in shape λ iff $(\star)$ fails, i.e. iff $s_\lambda = \sum F_{\mathrm{Des}T}$ contains F_A, F_B with $\varnothing \neq B \subseteq \operatorname{int} A$. Then twin existence would be read off the fundamental QSym expansion alone. Evidence: vacuous for $n \leq 13$ and $4/4$ at $n = 14$, which is weak because 14 is the first non-vacuous size. The separator is $n = 15$: list the violators and decide each one.

(2) **Mechanism (ii)**: non-twin failures, namely GW's $(4, 4, 1)$ and our H_3 type-5_2 cells, which fail without any twin (hunch). What replaces $e_z - e_x$ by sums such as $w_{14} + w_{44}$?

(3) **Uniform irreducible family in B_n**: Theorem B gives twins for all n, but an irreducible cell with a twin is computed only for $n \leq 5$. Neither natural w_J family is irreducible for every n.

Asks for Clio. (a) A hostile read of Lemma S, especially the S-ii coefficient bookkeeping, and of the §4 certificate logic. The reduction to four y's, the endpoint argument, and whether replaying an independent checker suffices as a proof of the four $\mu = 1$ are the points to attack. (b) Do you know anyone who owns type-A dominated twins (under any name), or a QSym/fundamental-expansion criterion for KL cones? Our owner checks of [GW] and [St] were negative for both.

References

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